Surja Sanyal

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Research Objectives

- Specializing in AI-driven cybersecurity and adaptive Intrusion Detection Systems for Industrial IoT networks.
- Driven to develop edge-tractable, poison-resistant defense frameworks for autonomous zero-day threat mitigation.
- Seeking a faculty position to establish a research program leveraging generative deep learning, game-theoretic optimization, and trustless collaboration expertise for real-world security and resource-aware deployment challenges.

Areas of Interest

- Generative Deep Learning
- Zero-Day Threat Mitigation
- Edge and Fog Computing
- AI Explainability and Reliability
- Spatio-Temporal Graphs
- Industrial IoT Security
- Collaborative IDS Defense
- Federated Learning Systems
- Continual Openset Learning
- Network Trust Management
- Game-Theoretic Optimization
- Adversarial Machine Learning

Academic Background

2021 – Present
- **Doctor of Philosophy (Ph.D.) — Computer Science and Engineering.**
  1. **Indian Institute of Technology Guwahati, Assam, India,** *jointly with*
  2. **Indian Institute of Technology (BHU) Varanasi, Uttar Pradesh, India.**

  CGPA: 8.75/10.00
  Thesis title: *Resource-Optimized Detection of Mimicry Attacks in Industrial IoT Networks Using Deep Learning Techniques.*

2019 – 2021
- **Master of Technology (M.Tech.) — Computer Science and Engineering.**
  National Institute of Technology Durgapur (NIT Durgapur), West Bengal, India.
  CGPA: 9.28/10.00
  Thesis title: *A Game Theoretic Framework for Surplus Food Distribution in Smart Cities and Beyond.*

2006 – 2010
- **Bachelor of Engineering (B.E.) — Computer Science and Engineering.**
  University Institute of Technology (UIT), West Bengal, India.
  Affiliated with The University of Burdwan (BU).
  Result: 79.9%

2006
- **CBSE AISSCE (Class XII).**
  Kendriya Vidyalaya Number 1, Saltlake, Kolkata, West Bengal, India.
  Subjects: Physics, Chemistry, Mathematics, Biology, English.
  Result: 79.6%

2004
- **CBSE AISSE (Class X).**
  Kendriya Vidyalaya Number 1, Saltlake, Kolkata, West Bengal, India.
  Result: 82.8%

Competitive Exams

TOEFL – 2021
- **Secured 107 marks out of a total 120**, conducted by the *Educational Testing Service (ETS).*

GATE – 2019
- **Secured 96.88 percentile score** in the subject *Computer Science and Information Technology*, conducted by the *Indian Institute of Technology (IIT) Madras.*

Research Experience

2021 – Present

- **Doctor of Philosophy (Ph.D.):** *IIT Guwahati* and *IIT (BHU) Varanasi*
  Focus: AI-Driven Cybersecurity & Adaptive IDS for IoT Networks
- Architected spatio-temporal graph diffusion ensembles utilizing GCN-GRU backbones to execute explainable, zero-day mimicry attack detection at the resource-constrained IoT edge.
- Engineered conjoint detection mechanisms via Multi-resolution Ring Buffers and random convolutional kernel transform (ROCKET) models, fusing host behavior with network context for continuous anomaly fingerprinting and $\mathcal{O}(1)$ swarm vaccinations.
- Designed lightweight, privacy-preserving collaborative IDSs using Bloom filters and distributed LSTM Autoencoders, ensuring robust consensus and adaptive scheduling even in majority-dishonest peer environments.
- Developed autonomous, zero-overhead trust management and adaptive placement algorithms, dynamically orchestrating heterogeneous deep learning ensembles (e.g., InceptionTime, Dilated TCN, Time Series Mixer) to maximize network lifetime.

2019 – 2021

- **Master of Technology (M.Tech.):** *NIT Durgapur*
  Focus: Game-Theoretic Frameworks & Smart City Optimization
- Formulated game-theoretic models and incentive mechanisms, balancing exploration and exploitation to optimize volunteer mobility and surplus food distribution in dynamic urban environments, leveraging available smart city platforms.
- Devised dynamic preference aggregation frameworks using bucketing algorithms and Shapley values to optimize crowdsourced project selection and enterprise decision-making.
- Enhanced intelligent urban transportation systems by optimizing curvelet entropy features for voice-assisted traffic sign recognition.

Research Publications

Published and Accepted Journal Articles

1. **Sanyal**, **S.**, Khatua, M., & Chattopadhyay, P. (2026). LITMUS: A Lightweight Collaborative Intrusion Detection System for Host Oriented Mimicry Attack Detection in Industrial Internet of Things Networks with Dishonest Collaborators in the Majority. *Computer Networks*, *286*, 112474–112492. **(IF: 4.7)**. doi:10.1016/j.comnet.2026.112474
2. **Sanyal**, **S.**, Khatua, M., & Chattopadhyay, P. (2026). CIPHER: A Collaborative IDS Placement and Scheduling Framework for Host-Oriented Mimicry Attack Detection in IoT Networks. *IEEE Internet of Things Journal*, *13*(14), 31880–31888. **(IF: 8.9)**. doi:10.1109/JIOT.2026.3689273
3. **Sanyal**, **S.**, Mukhopadhyay, S., & Xhafa, F. (2023). Enhanced Incentive Schemes for Volunteers to Distribute Surplus Food in Smart Cities. *Sustainable Cities and Society*, *91*, 104409. **(IF: 13.3)**. doi:10.1016/j.scs.2023.104409
4. **Sanyal**, **S.**, Kumar Singh, V., Xhafa, F., Sanyal, B., & Mukhopadhyay, S. (2021). A Game Theoretic Framework for Surplus Food Distribution in Smart Cities and Beyond. *Applied Sciences*, *11*(11), 5058. **(IF: 2.9)**. doi:10.3390/app11115058

Journal Articles Under Review and Revision

1. **Sanyal**, **S.**, Khatua, M., & Chattopadhyay, P. (2026). MIRAGE: An Adaptive Open-Set Spatio-Temporal Graph Diffusion Ensemble for Explainable Zero-Day Mimicry Detection at the IoT Edge. *Expert Systems With Applications*, 1–42.
2. **Sanyal**, **S.**, Khatua, M., & Chattopadhyay, P. (2026). HADRON: Autonomous Placement and Scheduling of Heterogeneous IDS using Silent Trust for Mimicry Attack Detection in IoT Networks. *Computer Networks*, 1–19.
3. **Sanyal**, **S.**, Khatua, M., & Chattopadhyay, P. (2026). JARVIS: Ultralight Conjoint Mimicry Attack Detection with Continuous Anomaly Fingerprinting and Fast Swarm Vaccinations in IoT Networks. *Internet of Things*, 1–24.

4. **Sanyal**, **S.**, Mukhopadhyay, S., Sanyal, B., & Mukhopadhyay, J. (2026). A Dynamic Preference Aggregation Mechanism using Shapley Scores and K-Means Clustering for Enterprise Decision Making in Smart Cities. *Knowledge and Information Systems*, 1–26.

5. **Sanyal**, **S.**, Khatua, M., & Chattopadhyay, P. (2026). SIPHON: A Parent Switch Indifferent Placement and Scheduling of Lightweight IDS for Mimicry Attack Detection in IoT Networks. *IEEE Transactions on Reliability*, 1–12.

6. Kumar, R., **Sanyal**, **S.**, Mukhopadhyay, S., & Xhafa, F. (2025). Volunteer Mobility Management for Surplus Food Distribution in Dynamic Environments in Smart Cities. *Sustainable Cities and Society*, 1–94. **(Under Revision)**.

Conference Proceedings

1. **Sanyal**, **S.**, Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2026, July). A Survey of Project Selection in Tourist Spots using Smart City Platforms. In *Proc. of the 20th International Conference on Complex, Intelligent, and Software Intensive Systems (CISIS-2026)* (pp. 122–134). doi:10.1007/978-3-032-29430-2_11

2. Sanyal, B., & **Sanyal**, **S.** (2026, June). Traffic Sign Recognition with Voice Assistance Using Optimized Curvelet Entropy Features. In *Proc. of the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). IEEE. doi:10.1109/GCON69192.2026.11648793

3. **Sanyal**, **S.**, Khatua, M., & Chattopadhyay, P. (2026, June). CSIDS: Detecting Mimicry Attacks using Network Context with Host Behavior in IoT Networks. In *Proc. of the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). IEEE. doi:10.1109/GCON69192.2026.11649178

4. **Sanyal**, **S.**, Priya, T., Debnath, N., Mitra, D., & Mukhopadhyay, S. (2026, June). Placing Tourist Spots in Smart Cities: A Crowdsourced Game Theoretic Approach. In *Proc. of the 3rd International Conference of the IEEE Guwahati Subsection (GCON 2026)* (pp. 1–6). IEEE. doi:10.1109/GCON69192.2026.11648713

5. **Sanyal**, **S.**, Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2026, May). FIREBOLT: Visitor Frequency Aware Crowdsourced Project Selection for Tourist Spots in Smart Cities using Bucketed Shapley Scores. In *Proc. of the 14th International Conference on Emerging Internet, Data & Web Technologies (EIDWT-2026)* (Vol. 291, pp. 255–266). Springer. doi:10.1007/978-3-032-22798-0_23

6. **Sanyal**, **S.**, Khatua, M., & Chattopadhyay, P. (2024, December). An Adaptive IDS Scheduling Framework for Host Based Mimicry Attack Detection in IoT Networks. In *Proc. of 2024 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS 2024)* (pp. 1–6). IEEE. doi:10.1109/ANTS63515.2024.10898203

7. **Sanyal**, **S.**, Mukhopadhyay, S., Xhafa, F., & Mukhopadhyay, J. (2024, June). Crowdsourced Project Selection in Tourist Spots with Voting in Smart Cities. In *Proc. of the Decision Science Alliance International Summer Conference (DSA ISC 2024)* (pp. 105–119). Springer. doi:10.1007/978-3-031-78241-1_10

8. **Sanyal**, **S.**, Mukhopadhyay, S., Xhafa, F., Singh, V. K., & Sanyal, B. (2022, August). An Incentive Framework for Surplus Food Distribution in the Context of Smart Cities. In *Proc. of the International Conference on Network-Based Information Systems (NBiS 2022)* (pp. 35–47). Springer. doi:10.1007/978-3-031-14314-4_4

Books and Chapters

1. **Sanyal**, **S.**, Mukhopadhyay, S., Xhafa, F., Kumar Singh, V., & Sanyal, B. (2023, May). Incentives in Surplus Food Distribution for Smart Cities and Beyond: An Activity Aware Solution. In *Advances in engineering and information science toward smart city and beyond* (pp. 75–90). doi:10.1007/978-3-031-29301-6_4

Technical and Literary Skills

Coding Languages	›	Python, Cython, C, SQL, VBA, SAS, and Bash/Shell scripting.
AI/ML Frameworks	›	PyTorch, PyTorch Geometric, TensorFlow, Keras, Scikit-Learn, NumPy, SciPy, and Pandas.
Tools and Systems	›	PARAM Supercomputers, SLURM, UNIX/Linux, tShark, VS Code, Arduino, and LaTeX.
Communication	›	Reading, writing, and speaking in English, Hindi, Bengali, Sanskrit, and Assamese.

Industry Experience

2011 – 2019 › **Bio-Statistical Programmer, Data Modeler, and Data Analyst.**
Tata Consultancy Services (TCS), Delhi-NCR, India.

1. I have worked for a Danish Pharmaceutical Client* in the Lifesciences domain as a SAS Biostatistical programmer doing data analysis and interpretation to establish the safety and efficacy of upcoming drugs undergoing clinical trials in phases 1, 2, and 3.
2. I have also worked for an American Insurance Client* in the Healthcare industry for health-related data modeling and trend prediction of US citizens.

* *Note:* Names of the client organizations have not been disclosed due to contractual obligations.

Academic Experience

Teaching Assistance

IIT Guwahati
- **Relational Database Management Systems** for **Coursera** from *September to November,* 2026, 2025, 2024.
- **Cloud Computing** for **Coursera** from *September to November,* 2026.
- **Database Management Systems** for the **Institute** from *January to May,* 2026.
- **Internet of Things** for the **Institute** from *July to November,* 2024, 2023, 2022.
- **Computer Systems** for the **Institute** from *January to May,* 2024.
- **Computer and Network Security** for the **Institute** from *January to May,* 2023.
- **Computer Networks Lab** for the **Institute** from *July to November,* 2022.
- **Topics in Networks** for the **Institute** from *January to May,* 2022.

IIT (BHU) Varanasi
- **Computer Networks** for the **Institute** from *July to November,* 2025.
- **Algorithms** for the **Institute** from *January to May,* 2025.

Mentorship Programs

IIT Guwahati
- Co-instructor in **One-week workshop** on **Power Electronics Hardware Design with Microcontroller Programming** from *June* 24 to *June* 30, 2023 sponsored by the *Science and Engineering Research Board (SERB), India* under the *Accelerate Vigyan scheme.*

Certifications and Trainings

High Performance Computing
- **Two-day workshop** on **High Performance Computing (HPC)** organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *Centre for Development of Advanced Computing (C-DAC), India* on *December* 2022.

Blockchain Applications
- **Three-day workshop** on **Scaling Blockchains: From Bitcoin to the Lightning Network** organized by the *Indian Institute of Technology Guwahati, India* in collaboration with the *University of Catania, Italy* on *August* 2022.

Certified Sanskrit Linguist
- **Diploma Degree** awarded by the *Central Sanskrit University (CSU), Delhi, India*, organized by the *Indian Knowledge Systems (IKS)* in association with the *Indian Institute of Technology Guwahati, India*, on *June* 2024.

Certified Karate Trainer
- **Second Dan Black Belt and Registered Trainer** awarded by the *Traditional Shito-Ryu Karate Do Association (India), TSKA*, approved by the *Karate India Organization (KIO)*, affiliated with the *World Karate Federation (WKF)*, recognized by the *International Olympic Committee (IOC)*, on *December* 2022.

Awards and Achievements

Track and Field	› Active participation awards in **Several competitive and charitable track and field events** from 2011 to 2020.
TCS Delhi-NCR	› Received **Service and Commitment** award on 2019, 2016, and 2014. › Received **Outstanding Contribution** award on 2017, 2016, and 2015. › Received **Distinguished Training** award on 2015, 2014, and 2013. › Received **Best Team of the Year** award on 2011.

References

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